

Analyzing Civilian Casualty Rates and Uncertainty in Yemen Drone Strikes

Salvatore Cayabyab, Ziteng Cheng, Wonhyeok Choi, Katinka Lennemann, & Kwanhee Yoon

STATS140XP Statistical Consulting, UCLA
Email: klennemann@ucla.edu

ABSTRACT

This report examines civilian casualties from U.S. strikes in Yemen, addressing whether drone or non-drone strikes result in lower civilian casualty rates and whether reporting uncertainty has changed over time. Using Bureau of Investigative Journalism data from 2002-2016, we analyzed 324 strikes through Wilcoxon rank-sum tests and linear regression. While drone strikes showed lower average civilian casualties (mean minimum: 0.411 vs. 1.382), the difference was not statistically significant ($p=0.7842$). However, uncertainty in casualty reporting, measured as the difference between minimum and maximum deaths, significantly decreased over time ($p=0.011$). These findings indicate that drone strikes do not demonstrate significantly lower civilian casualties, though data collection precision has improved.

Key words: Drones, civilian, casualties, Yemen

1 INTRODUCTION

The U.S. has been carrying out strikes in Yemen as a part of counter terrorism efforts, in addition to also carrying out strikes in Afghanistan, Somalia, and Pakistan. These covert operations often are not accompanied by transparent data releases or disclosures. Therefore, there exists much uncertainty about the true efficacy and price of using drone strikes with respect to civilian casualties. This report aims to provide a statistical analysis of civilian casualty rates with respect to drone strikes, and also how uncertainty of these casualty estimates may be changing over time.

2 BACKGROUND

2.1 Literature Review

Lewis and Vavrichuk (2017) published a book called *Rethinking the Drone War*, which discusses drone strike casualties over time. Analysis of Pakistan civilian casualty rates for instance is provided in this book, but was not illuminated for other countries. Nor were drone strike casualties situated within the context of drone versus non-drone strikes, which can shed important light on what the cost of counter-terrorism strikes might look like in multiple different operation modes.

Other existing literature also discusses the ethicality and legality of drone strikes, such as Keene (2015) from U.S. Army War College, but without the backing of numbers to situate what collateral damage numbers might truly look like.

Furthermore, the numbers for casualty rates often are associated with much uncertainty. An article by ABC News (2011) discusses how the U.S. may be potentially under-reporting civilian casualties from drone strikes. Human Rights Clinic (2012) has also published scholarship on how numbers from tracking organizations that report drone strike deaths have various methodological limitations, and Woods (2017) from the Bureau of Investigative Journalism points out inconsistencies in reported numbers by a big-name media outlet.

Therefore, this paper aims to highlight important context of how costly drone strikes are compared to non-drone strikes with respect to civilian casualty rates. Furthermore, this report examines the uncertainty for reported civilian casualty numbers over time to see if reporting is getting better or not. This paper uses data from a dataset from the Bureau of Investigative Journalism to carry out analysis (dro (Nd)).

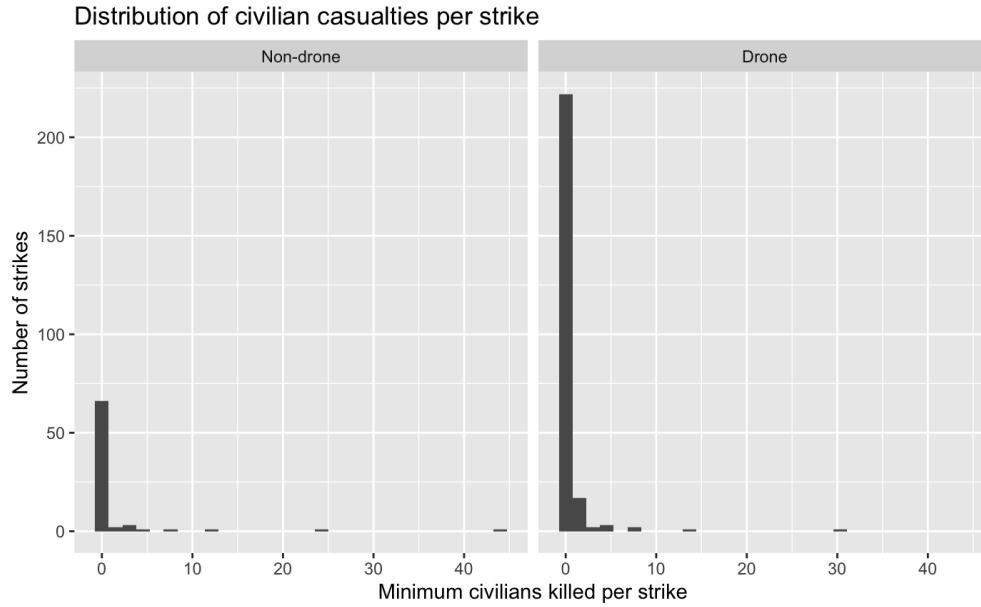


Figure 1. Distribution of civilian casualties

In other words, this paper explores the two following research questions:

- (i) In Yemen, do drone strikes have a lower civilian casualty rate per strike than non-drone strikes?
- (ii) Did casualty numbers become more precise (less uncertainty) over time in Yemen?

3 METHODS

3.1 Exploratory Data Analysis

This research paper focuses on the following variables:

- **strike_type:** a binary column coded as a 0 or 1, with 1 indicating that this observation was a drone strike (as opposed to a general air strike, ground operation, or missile strike). The proportion of drone strikes to non-drone strikes in the data is 76 to 248, indicating a class imbalance.
- **date:** Month, day, and year date of the strike's occurrence, which ranges from 2002-2019. There lies a seven year gap between the first reported drone strike in 2002 and the next subsequent one in 2009, making the 2002 drone strike an outlier.
- **min_civ:** The minimum number of civilians killed, based on sources at each strike
- **max_civ:** The maximum number of civilians killed based on sources at each strike
- **diff_civ:** A new column added to the dataset which is calculated as the difference between the maximum and minimum number killed. This value can be used as a measure of uncertainty for civilian casualty rates.

For the first research question, the `strike_type` column is used as a binary predictor variable for the outcome of either the minimum or maximum number of civilian killed. For the second research question, the relationship is explored between the date as the independent variable and the difference between min and max civilian casualties as the dependent variable.

The distribution of minimum civilian casualty rates is visualized in the two histograms in Figure 1, demonstrating that these values are not normally distributed. Maximum civilian casualty rates were found to be similarly distributed on a skewed scale.

Moreover, the means of the minimum and maximum civilian casualties by strike type are also shown in the table to the right. Whether looking at the minimum or maximum number of civilians killed, non-drone strikes have a higher mean value than drone strikes, lending support to the idea that drone strikes harm less civilians.

	Non-drone strike	Drone strike
Avg. min killed	1.382	0.411
Avg. max killed	1.763	0.770

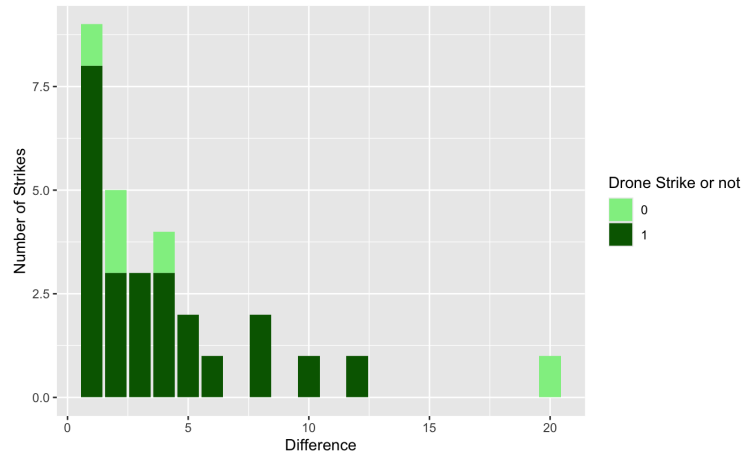


Figure 2. Non-zero differences in min and max civilian casualties by strike type

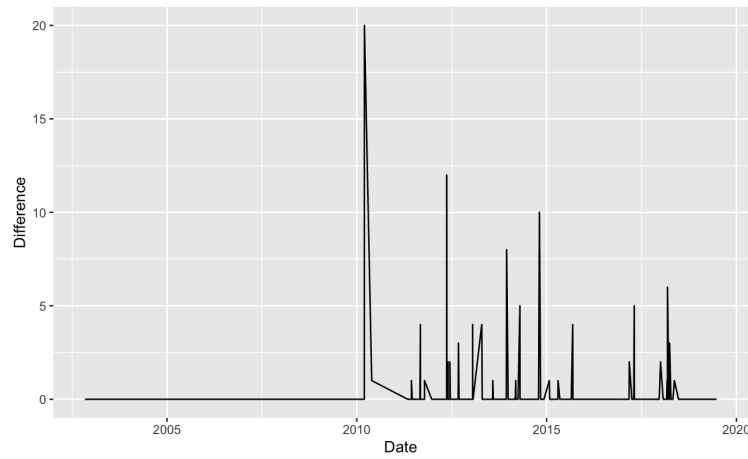


Figure 3. Difference in min and max civilian casualties over time

In addition, the percent of strikes where the difference between min and max killed was 0 turns out to be 91%. In other words, the reported minimum and maximum civilian counts match in 91% of strike observations.

For the strikes where the difference in minimum and maximum values wasn't zero, those observations were plotted in Figure 2. This chart demonstrates that for non-zero differences, the values ranged between 1-20. The stacked bars indicating strike type indicate that most of the drone strikes have smaller differences in casualty min/max values, save for the outlier value of 20.

Lastly, difference in drone strike reporting can also be plotted over time, as shown in Figure 3. Generally, the difference seems to be shrinking over time, suggesting greater precision in reporting over time.

3.2 Research questions

With a better understanding of the variables at hand, the two following null and alternate were used to test the first research question:

H_0 : Drone and non-drone strikes have the same civilian casualty rates.

H_1 : Drone strikes have a lower civilian casualty rate than non-drone strikes.

From the graphs above, it is evident that the civilian casualty rate is non-normally distributed, regardless if the minimum or the maximum reported value is used. Therefore, the nonparametric Wilcoxon rank-sum test was used to compare the values of these two populations at a significance level of $\alpha = 0.05$, for the $n = 324$ strikes. The test was run for both the minimum and the maximum civilian casualty rates to compare non-drone and drone strikes casualty occurrences.

For the second research question, the difference between the minimum and maximum number of total killed and number of civilians killed were used to evaluate uncertainty. These were plotted over time and aggregated by year in order to get a better idea of how death toll reporting uncertainty was changing over time. Independence of strikes from each other over time could not be established, so strikes were grouped into an early period (before 2014) and a late period (after 2014). Since the distribution of uncertainty values was not normal, a one-sided Wilcoxon ranked-sum test was used to test the following hypotheses:

H_0 : There is no difference between the strike uncertainty in the early period (2002-2014) versus the late period (2014-2019).

H_1 : Strike uncertainty in the early period (2002-2014) was significantly higher than in the late period (2014-2019).

The year 2014 was chosen as it is a midpoint between the years of the 2009 and 2019, which is the data range that the majority of the data spans, save for one drone strike in 2002.

4 RESULTS

The average minimum civilians killed values for drone and non-drone strikes were 0.411 and 1.382 respectively; however, the distributions of the two groups did not differ significantly (Wilcoxon ranked sum $W = 9730$, $n_1 = 76$, $n_2 = 248$, $p = 0.7842$).

The average maximum civilians killed values for drone and non-drone strikes were 1.763 and 0.770 respectively; the distributions of these two groups also did not differ significantly (Wilcoxon ranked sum $W = 9631$, $n_1 = 76$, $n_2 = 248$, $p = 0.6753$).

In other words, the results two Wilcoxon ranked-sum tests indicate that civilian casualty rate does not differ significantly for drone versus non-drone strikes, so there is not enough evidence to reject the null hypothesis.

With regards to the question of uncertainty, Figure 4 shows the plots of civilian casualties over time, this time with the mean values of the early versus the late period plotted as horizontal lines, and a dotted line for the 2014 divide between periods. Due to the discrete values of civilian casualties, a small amount of variation was added to each data point to more clearly demonstrate the distribution of values. The Wilcoxon ranked-sum test found that uncertainty in civilian casualty was significantly higher for the early period than the late ($W = 15824$, $p = 0.0001226$). Therefore, there is evidence to support the alternate hypothesis for research question 2 that in 2014-2019, uncertainty is significantly lower than in 2002-2014.

5 CONCLUSIONS

Summarizing casualty data by drone versus non-drone strikes indicate that drone strikes have on average lower civilian casualties in the current dataset of strike observations. However, this paper found that this difference was not significant enough to dispel the null hypothesis that strike type makes a difference. This finding is significant because proponent of drone strikes often tout the idea

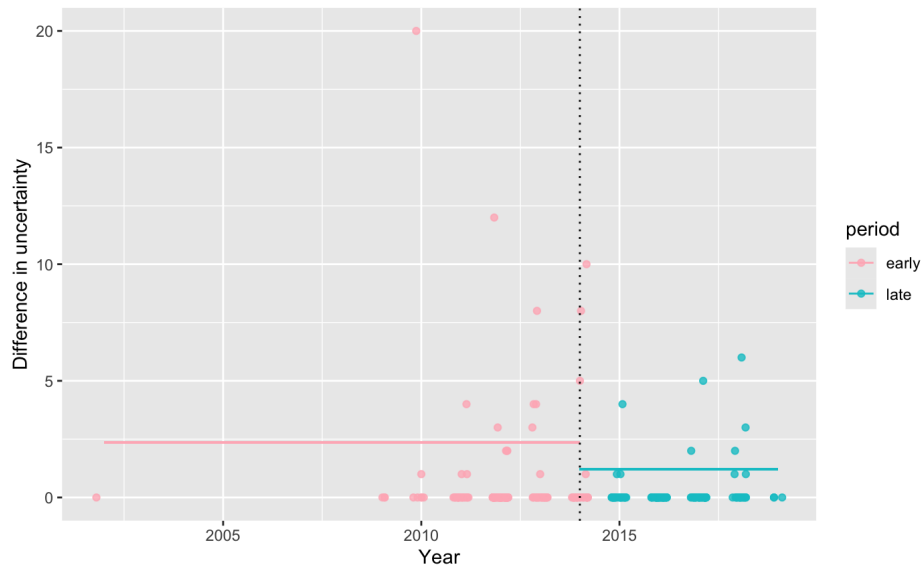


Figure 4. Uncertainty in Civilian Casualties over Time

that drone strikes are more targeted than non-drone strike operations. From this dataset on Yemen drone strikes, there is no support to the idea that drone strikes incur significantly lower the civilian casualties.

This paper also found that uncertainty in civilian casualty reporting from 2002 to 2019 has changed over time. In particular, the period of 2014-2019 has a significantly lower amount of uncertainty than the early period of 2002-2014. While previous literature has highlighted problems with uncertainty in reported data values, this report finds evidence to support the idea that at the very least, data collection by the Bureau of Investigative Journalism is on civilian casualty reporting is becoming more precise over time.

5.1 Limitations

This analysis is in part limited by the fact that data on drone strike versus non drone strikes have uneven class imbalances. The smaller amount of non-drone strike observations make assessing true class differences more difficult. Future research might consider carrying out analysis on a more complete dataset, or see if the trends found in this paper are still being observed. A change in presidential administration in 2020 may have also changed the accessibility of strike data or information sharing; additional research may take this into account during analysis.

Furthermore, this paper is limited in that it only examines strike data from Yemen. Therefore, more research should perhaps be done to analyze whether these findings hold true for other countries in which the U.S. carries out strike operations, such as Somalia, Afghanistan, or Pakistan, in order to see if drone strikes truly are less effective and that civilian casualty reporting is in fact becoming more precise.

While this report may also state that civilian casualty reporting has seen a trend in becoming more precise, elucidating why exactly this trend is occurring would also be valuable in understanding how to develop more transparent data reporting practices.

REFERENCES

- N.d., Drone warfare: The Bureau of Investigative Journalism.
- Humans Rights Clinic, T., 2012, Counting drone strike deaths: Technical report, Columbia Law School.
- Keene, S. D., 2015, Lethal and legal? the ethics of drone strikes: Technical report, U.S. Army War College Carlisle United States.
- Lewis, L., and D. M. Vavrich, 2017, Rethinking the drone war: National security, legitimacy and civilian casualties in u.s. counterterrorism operations: Marine Corps Association.
- Woods, C., 2017, Analysis: Cnn expert's civilian drone death numbers don't add up: The Bureau of Investigative Journalism.